

INITIAL ANALYSIS REPORT

HealthConnect Appointment Attendance and No-Show Analysis

Dataset: Health Connect Appointment Data

By

Ibtihaj Shuaib

Programme: Data Analytics Internship, AnalystLab Africa

August, 2026

CONTENTS

PROJECT AND RESOURCE REVIEW	3
Problem to Be Addressed.....	3
My Role in the HealthConnect Project	3
Resources to Be Used	4
Initial Approach	4
INITIAL ANALYSIS.....	5
Dataset and Data Dictionary Review	5
Initial Data Quality Assessment.....	6
Important Variables for Appointment Attendance and No-Show Analysis	8
Business Questions	9
Potential Key Performance Indicators (KPIs).....	10
Initial Analysis Approach	11
ASSUMPTIONS, LIMITATIONS, RISKS AND DEPENDENCIES	13
Assumptions:.....	13
Limitations:	13
Risks:.....	14
Dependencies:	14
CONCLUSION	15

PROJECT AND RESOURCE REVIEW

Problem to Be Addressed

HealthConnect Clinic is experiencing missed patient appointments, which can result in underutilised appointment slots and challenges in managing clinic resources effectively. The clinic also needs a clearer understanding of the factors and patterns associated with appointment no-shows.

This analysis will examine the available appointment data to understand attendance and no-show patterns and identify factors that may be associated with missed appointments. The findings will provide a data-driven basis for supporting efforts to improve appointment attendance, patient engagement, and appointment management.

The broader project question is: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

My Role in the HealthConnect Project

As a Data Analytics Intern, my specific focus will be to investigate the appointment data to understand patterns and factors associated with patient attendance and no-shows. I will contribute by examining the available appointment data to help the project team understand:

- Patterns in patient appointment attendance and no-shows.
- Variables that may be associated with missed appointments.
- Areas where data can support better appointment management.
- Key measures that can be used to monitor appointment attendance and no-shows.

Resources to Be Used

The main resources for my Data Analytics work are:

1. HealthConnect Appointment Dataset (HealthConnect_Appointment_Data.csv): This dataset contains fictional and anonymised appointment records.
2. HealthConnect Data Dictionary (HealthConnect_Data_Dictionary.xlsx): The Data Dictionary provides descriptions and explanations of the variables contained in the appointment dataset.
3. Data Analytics Tools: Appropriate data analytics tools such as SQL, Python, Tableau, and Power BI may be used during the project.

Initial Approach

My initial approach to the HealthConnect Data Analytics project will be to:

- ❖ Understand the data.
- ❖ Assess the data quality.
- ❖ Identify variables relevant to appointment attendance and no-shows.
- ❖ Develop business questions and propose relevant KPIs to guide the analysis.
- ❖ Conduct further analysis in subsequent stages to identify patterns and generate actionable business insights.

INITIAL ANALYSIS

Dataset and Data Dictionary Review

The HealthConnect Appointment Dataset was reviewed together with the HealthConnect Data Dictionary to understand the structure, content, and purpose of the available data before conducting further analysis.

The dataset contains 5,000 appointment records and 18 variables. Each record represents a scheduled patient appointment. The dataset includes information relating to patient demographics, appointment characteristics, booking information, previous appointment history, reminder activities, access and operational factors, and the final appointment outcome.

The Data Dictionary was used to understand the meaning and expected data type of each variable. It identifies `appointment_id` as the unique appointment identifier and `patient_id` as an anonymized patient identifier that may occur across multiple appointments.

Dataset Structure

The variables in the dataset can be grouped into the following areas:

Category	Variables
Patient information	<code>patient_id</code> , <code>gender</code> , <code>age</code> , <code>age_group</code>
Appointment information	<code>appointment_id</code> , <code>appointment_type</code> , <code>appointment_date</code> , <code>appointment_day</code> , <code>appointment_time</code>
Booking information	<code>booking_date</code> , <code>booking_lead_days</code>
Previous appointment history	<code>previous_appointments</code> , <code>previous_no_shows</code>

Reminder information	reminder_sent, reminder_channel
Access and operational factors	distance_to_clinic_km, waiting_time_minutes
Appointment outcome	appointment_outcome

The appointment_outcome variable represents the final status of an appointment and contains three possible outcomes: Attended, No-Show, and Cancelled.

The dataset therefore provides information that can be used to examine appointment outcomes in relation to patient, appointment, booking, reminder, and accessibility characteristics.

Initial Data Quality Assessment

An initial data quality assessment was conducted in Power Query to determine whether the dataset was sufficiently structured for the proposed analysis.

Missing Values

The distance_to_clinic_km variable contains 90 missing records (1.8%), while waiting_time_minutes contains 60 missing records (1.2%). Both are consistent with the Data Dictionary's indication of limited missing values. No missing values were identified in reminder_channel. The value None is treated as a valid category when no reminder was sent. The identified missing values were retained for consideration during subsequent analysis.

Duplicate Records Assessment

The dataset was checked for completely duplicated rows. After the duplicate check, the dataset remained at 5,000 records, indicating that no completely duplicated rows were identified.

The appointment_id variable was then checked separately because it is identified in the Data Dictionary as the unique appointment identifier and primary key. The dataset contained 5,000 unique appointment IDs across 5,000 records. Therefore, no duplicate appointment IDs were identified.

Data Type Validation

The data types of all variables were reviewed against the HealthConnect Data Dictionary. Two discrepancies were identified. The booking_date and appointment_date columns were initially stored as Text, whereas the Data Dictionary specifies them as Date fields. Both columns were converted from Text to Date using the appropriate locale setting.

The initial conversion using the Nigeria locale resulted in date conversion errors. After selecting the English (United States) locale, both date columns were successfully converted to the Date data type, with no remaining errors.

All other variables were found to have data types consistent with the Data Dictionary and did not require modification.

Basic Validity and Consistency Check

A basic validity and consistency review was conducted on key numerical variables to identify obvious values that could affect the planned analysis.

The age variable ranged from 18 to 80 years, consistent with the Data Dictionary's specification that the dataset contains adult patients only. booking_lead_days ranged from 0 to 60 days, with no negative values. previous_appointments ranged from 0 to 11, while previous_no_shows

ranged from 0 to 5. The Data Dictionary specifies that `previous_no_shows` should not exceed `previous_appointments`, and this relationship will be considered during further analysis.

Non-null values for `distance_to_clinic_km` ranged from 0.5 to 45 km, while `waiting_time_minutes` ranged from 2 to 64 minutes.

Overall, no obvious invalid or implausible values requiring correction were identified.

Categorical Consistency Check

The categorical variables, `gender`, `age_group`, `appointment_type`, `appointment_day`, `reminder_sent`, `reminder_channel`, and `appointment_outcome`, were reviewed against the HealthConnect Data Dictionary.

All observed categories aligned with the expected values. No unexpected categories, inconsistent labels, or obvious formatting inconsistencies were identified.

Important Variables for Appointment Attendance and No-Show Analysis

Based on the review of the Data Dictionary and the project objective, seven variables were identified as particularly relevant for investigating appointment attendance and no-show patterns:

1. **Appointment_outcome:** This is the primary outcome variable. It indicates the outcome of each appointment and will be used to examine attendance and no-show patterns.
2. **Booking_lead_days:** This represents the number of days between booking and the scheduled appointment. It can help investigate whether the length of time between booking and the appointment is associated with no-shows.

3. **Previous_no_shows:** This represents a patient's history of previously missed appointments. It can help determine whether previous no-show behaviour is associated with future no-shows.
4. **Appointment_type:** This identifies the type of appointment and can help determine whether attendance patterns differ across different appointment types.
5. **Reminder_sent:** This indicates whether a patient received an appointment reminder. It can help assess whether sending reminders is associated with improved appointment attendance.
6. **Distance_to_clinic_km:** This represents the distance between the patient and the clinic. It can help investigate whether accessibility may influence the likelihood of attending an appointment.
7. **Age_group:** This categorizes patients by age and can help identify whether appointment attendance and no-show patterns vary across different age groups.

These variables will form the basis for the business questions and subsequent analysis.

Business Questions

The analysis will seek to answer the following business questions:

1. Which characteristics of patients and appointments are linked to a higher likelihood of no-shows?

This will provide an overall understanding of the characteristics and circumstances associated with missed appointments.

2. Is a patient's previous no-show history associated with whether they miss a future appointment?

This will help determine whether previous no-show behaviour can be used to identify patients who may be at higher risk of missing future appointments.

3. Is the number of days between booking and the appointment associated with appointment no-shows?

This will help assess whether longer booking lead times may contribute to missed appointments.

4. Is sending an appointment reminder associated with improved appointment attendance?

This will help evaluate whether the reminder process may be useful in reducing missed appointments.

5. Do appointment no-show patterns differ by appointment type, patient age group, or distance to the clinic?

This will help identify patient or appointment characteristics that may be associated with differences in attendance behaviour.

Potential Key Performance Indicators (KPIs)

The following four KPIs are identified to support the investigation of appointment attendance and no-show patterns. These KPIs are proposed for use in subsequent stages of the project.

- 1. No-Show Rate:** This KPI will measure the percentage of scheduled appointments that result in a no-show. It provides a direct measure of the scale of the missed-appointment problem.

2. **Attendance Rate:** This KPI will measure the percentage of scheduled appointments that are attended. It provides an overall measure of appointment attendance and complements the No-Show Rate.
3. **Repeat No-Show Rate:** This KPI will measure the proportion of patients with a history of previous no-shows who subsequently miss another appointment. It will help assess whether previous no-show behaviour is associated with future appointment attendance.
4. **Reminder Attendance Rate:** This KPI will measure the percentage of appointments attended among patients who received an appointment reminder. It will help assess the relationship between reminder delivery and appointment attendance.

Initial Analysis Approach

1. **Analyze Appointment Outcomes:** The analysis will begin with the validated HealthConnect appointment dataset, using "appointment_outcome" as the primary outcome variable for investigating appointment attendance and no-show patterns.
2. **Examine Overall Attendance and No-Show Patterns:** The analysis will involve examining overall appointment outcomes to establish the general attendance and no-show pattern.
3. **Investigate Factors Associated with No-Shows:** The analysis will investigate how no-show patterns vary across key patient, appointment, booking, reminder, and accessibility factors.
4. **Calculate and Evaluate the Proposed KPIs:** The four proposed KPIs will be calculated and evaluated during the subsequent stages of the project. These measures will provide a basis for assessing appointment performance and investigating the identified business questions.

5. Compare Relevant Groups and Identify Patterns: The analysis will then compare relevant groups and identify patterns that may help explain differences in appointment attendance. Particular attention will be given to previous no-show behaviour, reminder delivery, booking lead time, appointment type, patient age group, and distance to the clinic.
6. Develop Business Insights and Recommendations: Finally, the findings will be translated into business insights and presented using appropriate visualizations. The objective will be to identify potential areas where HealthConnect could improve appointment attendance and reduce no-shows.

ASSUMPTIONS, LIMITATIONS, RISKS AND DEPENDENCIES

The following factors were identified as considerations that may affect the proposed approach to analysing appointment attendance and no-show patterns.

Assumptions:

- The appointment_outcome field accurately represents the recorded outcome of each appointment and can therefore be used as the primary outcome variable.
- The variables selected for the analysis are appropriately defined and represented according to the HealthConnect Data Dictionary.
- Booking_lead_days accurately represents the number of days between booking and the scheduled appointment.
- Previous_no_shows accurately represents a patient's historical no-show behaviour.

Limitations:

- Some variables contain missing values, including limited missing information in distance_to_clinic_km and waiting_time_minutes.
- The dataset does not capture every possible reason why a patient may miss an appointment, such as illness, emergencies, transportation difficulties, or changes in personal circumstances.
- The available variables can help identify relationships and patterns but may not be sufficient to establish causal relationships between a factor and appointment no-shows.
- The dataset is fictional and anonymised; therefore, findings may not necessarily represent actual patient behaviour in a real healthcare setting.

Risks:

- Misinterpreting associations between variables as evidence of causation could lead to inappropriate conclusions or recommendations.
- Missing or incomplete information may affect comparisons between patient or appointment groups.
- Incorrect interpretation of variables or inconsistent use of definitions could affect the validity of the analysis.

Dependencies:

- The analysis depends on the HealthConnect Data Dictionary for the correct interpretation and use of dataset variables.
- The validated dataset is required before the proposed KPIs can be calculated and analysed.
- The accuracy and consistency of appointment_outcome are particularly important because it serves as the primary outcome variable.
- Subsequent analysis will depend on the availability of appropriate analytical and visualization tools.

CONCLUSION

The Week 4 review established a clear foundation for analysing appointment attendance and no-show patterns at HealthConnect Clinic. The dataset and Data Dictionary were reviewed, and the initial data-quality assessment identified and addressed the main data-type issue while confirming that the dataset contained no duplicate records or obvious invalid values.

The review also identified key variables, business questions, and potential KPIs that will guide the subsequent analysis. While missing information and other limitations may affect the analysis, these considerations have been identified and will be taken into account when interpreting the findings.

Overall, the proposed approach provides a structured basis for investigating appointment no-shows and generating data-driven insights to support improved appointment attendance and appointment management in subsequent stages of the project.